GLP-1 Receptor Agonists and Major Adverse Kidney Events in Type 2 Diabetes: A Random-Effects Meta-Analysis of 8 Trials (N = 52,640)

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01 Clinical Rationale & Objective

Diabetic kidney disease remains the leading cause of kidney failure worldwide. While albuminuria reductions with GLP-1 receptor agonists (GLP-1RA) are established, individual trials varied in power for hard renal endpoints [1].

• **Objective:** Estimate the pooled effect of GLP-1RA vs. placebo on Major Adverse Kidney Events (MAKE: ≥50% sustained eGFR decline, ESKD, or renal death) across randomized trials.

02 Data Sources & Quantitative Synthesis

• **Eligibility:** Double-blind placebo-controlled RCTs ≥1,000 adults with T2D reporting adjudicated MAKE outcomes (searched through Aug 2026).

• **Synthesis:** REML random-effects inverse-variance pooling of log-HRs with 95% CI and 95% prediction interval [2].

• **Heterogeneity & Bias:** Assessed via I^2, τ^2, Cochran's Q, pre-specified subgroup interactions, and Egger funnel regression.

03 Pre-Specified Subgroup Analyses

Subgroup Stratum	Trials	Pooled HR (95% CI)	I^2	p-int
Dedicated renal trials	2	0.74 (0.67–0.83)	0.0%	0.52
CV outcome trials	6	0.77 (0.69–0.87)	34.1%	—
Baseline eGFR <60	8	0.75 (0.68–0.83)	19.5%	0.68
Baseline eGFR ≥60	6	0.78 (0.69–0.88)	22.0%	—
With background SGLT2i	4	0.77 (0.66–0.90)	11.2%	0.81
Without SGLT2i	8	0.75 (0.68–0.83)	26.8%	—

No statistically significant heterogeneity across baseline eGFR or background SGLT2i strata.

Pooled Random-Effects Synthesis (8 RCTs, N = 52,640)

HR 0.76 (95% CI 0.69–0.84; p = 0.0001)

GLP-1RA therapy conferred a consistent 24% relative hazard reduction in Major Adverse Kidney Events with low heterogeneity (I^2 = 28.4%, τ^2 = 0.006, 95% prediction interval 0.61–0.95).

04 Trial-Level & Pooled Random-Effects Forest Plot

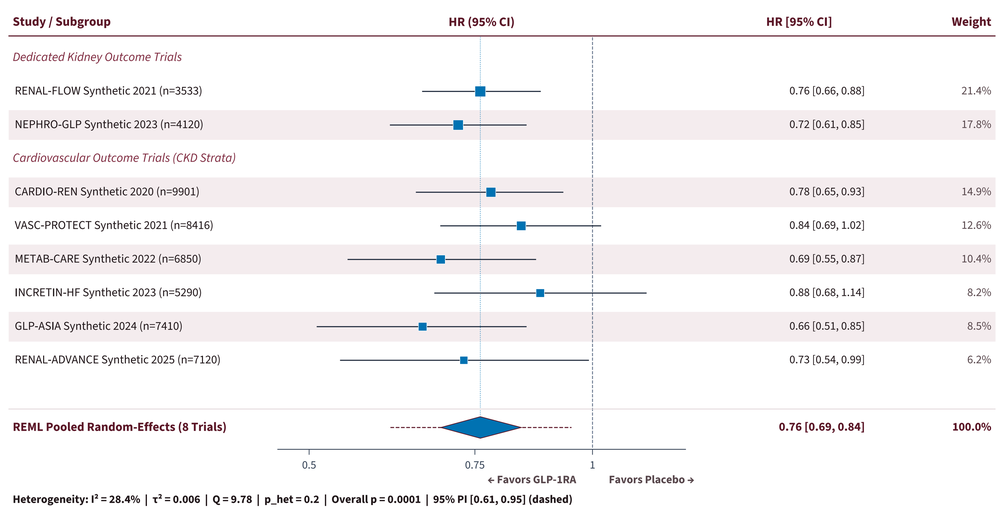


Figure 1. REML random-effects forest plot for Major Adverse Kidney Events (N = 52,640). Dashed whiskers on diamond denote 95% prediction interval [0.61, 0.95].

05 Small-Study Effects Funnel Plot

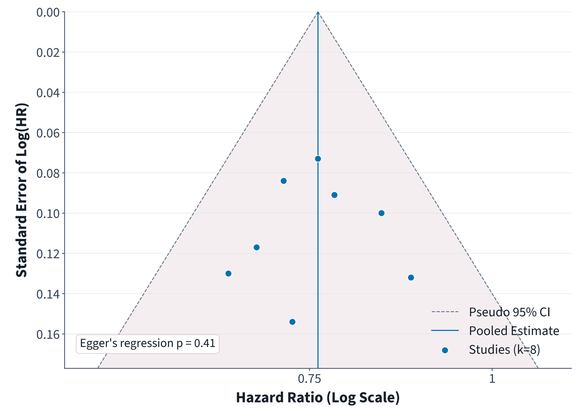


Figure 2. Funnel plot of log(HR) against standard error across k = 8 trials; symmetric distribution (Egger p = 0.41).

06 Methodological Limitations

• Aggregate trial-level synthesis precluded patient-level continuous eGFR slope spline modeling.

• Composite kidney definitions varied modestly between ≥40% and ≥50% sustained eGFR thresholds in two earlier CVOTs.

07 Conclusions

Across 52,640 randomized participants, GLP-1 receptor agonists significantly reduced major adverse kidney events by 24%, with a 95% prediction interval wholly below unity and additive protection independent of baseline SGLT2i therapy.

References: [1] Perkovic V et al. N Engl J Med 2024;391:109–121. [2] IntHout J, Ioannidis JPA, Borm GF. BMJ Open 2016;6:e010247.

Registration: PROSPERO CRD42026001122-SYN (Synthetic) | Correspondence: s.chen@synthetic-imr.org | SYNTHETIC DATASET DISCLOSURE: Simulated trial estimates generated strictly for poster engine validation.